

BLUESTOCK MUTUAL FUND ANALYTICS

Final Capstone Project Report

Data-Driven Analysis of Mutual Fund Performance, Risk and Investor Behaviour

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1.EXECUTIVE SUMMARY

The Bluestock Mutual Fund Analytics project was developed to provide a data-driven analysis of mutual fund performance, risk, investor behaviour and portfolio concentration.

The project integrates multiple datasets covering fund master information, historical Net Asset Value (NAV), assets under management, SIP inflows, category-level inflows, investor transactions, portfolio holdings, scheme performance and benchmark indices.

A structured analytical workflow was developed using Python and Pandas to load, clean and transform the datasets. Historical NAV data was used to calculate daily returns and risk-adjusted performance measures. SQL and SQLite were used for structured data storage and querying, while Power BI was used to develop an interactive dashboard.

The analysis was extended through advanced analytics including Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), rolling 90-day Sharpe ratio, investor cohort analysis, SIP continuity analysis, a simple risk-based fund recommender and sector-level Herfindahl-Hirschman Index (HHI) concentration.

The project analysed **40 mutual fund schemes across 10 fund houses**. The results provide insights into fund performance, downside risk, investor contribution patterns, SIP behaviour and portfolio concentration.

The final solution combines data engineering, financial analytics, statistical analysis and interactive visualization into a single mutual fund analytics framework.

2.INTRODUCTION

Mutual funds provide investors with access to professionally managed investment portfolios across different asset classes and categories. They are widely used by investors because they provide diversification and professional management.

However, selecting and evaluating mutual funds requires more than simply comparing historical returns. Investors need to consider risk, volatility, downside exposure, portfolio concentration, fund category, investment behaviour and consistency of performance.

The Bluestock Mutual Fund Analytics project was developed to address this requirement by combining multiple mutual fund datasets into a structured analytical solution.

The project uses Python for data processing and statistical analysis, SQLite for structured data storage, Jupyter Notebook for exploratory and advanced analytics, and Power BI for interactive dashboard development.

The project follows the complete analytical lifecycle from raw data ingestion and cleaning to performance analysis, risk measurement, investor behaviour analysis and final visualization.

3.PROBLEM STATEMENT

The project contains multiple datasets related to mutual funds, including fund master information, historical NAV, AUM, SIP inflows, investor transactions, portfolio holdings and benchmark indices.

When these datasets are analysed independently, it becomes difficult to obtain a complete understanding of fund performance and investor behaviour.

The main objective of this project is therefore to develop a structured analytical workflow that transforms the available raw datasets into meaningful insights.

The project aims to answer questions such as:

- Which mutual funds demonstrate stronger risk-adjusted performance?
- Which funds have higher historical downside risk?
- Which investor cohorts contribute the highest investment amounts?
- How regularly do investors continue their SIP investments?
- Which funds have higher portfolio concentration?
- Which funds can be recommended for different risk appetites?
- How can these findings be presented through an interactive dashboard?

4.PROJECT OBJECTIVES

The major objectives of the project are:

1. To ingest and organize the available mutual fund datasets.
2. To clean and standardize the datasets.
3. To validate fund and AMFI code information.
4. To store structured data using SQLite.
5. To perform exploratory data analysis.
6. To analyse historical mutual fund performance.
7. To calculate risk metrics such as VaR and CVaR.
8. To calculate rolling Sharpe ratios.
9. To analyse investor cohorts.
- 10.To analyse SIP continuity and identify at-risk investors.
- 11.To develop a simple risk-based fund recommender.
- 12.To measure sector concentration using HHI.
- 13.To develop an interactive Power BI dashboard.
- 14.To generate business insights and recommendations.

5.DATA SOURCES

The project uses ten major datasets.

Dataset	Purpose
Fund Master	Scheme, fund house, category, plan and risk information
NAV History	Historical daily NAV and return calculations
AUM by Fund House	Fund-house-level AUM analysis
Monthly SIP Inflows	Monthly SIP trend analysis
Category Inflows	Category-wise investment inflows
Industry Folio Count	Folio growth analysis
Scheme Performance	Mutual fund performance metrics
Investor Transactions	Investor and SIP behaviour analysis
Portfolio Holdings	Sector allocation and concentration analysis
Benchmark Indices	Benchmark comparison

The Fund Master dataset contains **40 schemes and 15 columns**. Important Fund Master fields include:

- AMFI code
- Fund house
- Scheme name
- Category, Sub-category
- Plan
- Launch date
- Benchmark
- Expense ratio
- Exit load
- Minimum SIP amount, Minimum lump-sum amount
- Fund manager
- Risk category, SEBI category code

The NAV History dataset contains approximately **46,000 NAV observations** covering 40 schemes. The available NAV history ranges from : **3 January 2022 to 29 May 2026**.

6. ETL DESIGN AND ARCHITECTURE

The project follows an Extract, Transform and Load architecture.

6.1 Extract

The extraction stage involves loading the available CSV datasets into Python using Pandas. The project also supports live NAV retrieval where required. The raw datasets are stored separately from processed datasets to maintain a clear distinction between original and cleaned data.

6.2 Transform

The transformation stage includes:

- Standardizing column names
- Converting dates into datetime format
- Converting numeric columns to appropriate data types
- Checking missing values
- Checking duplicate records
- Validating AMFI codes
- Sorting NAV data by scheme and date
- Calculating daily returns
- Preparing datasets for analysis
- Creating analytical metrics

6.3 Load

After cleaning and transformation, the datasets are stored in processed form and loaded into SQLite where required. The structured data can then be queried using SQL and consumed by Python analytics and Power BI.

ETL Flow

Raw CSV Data



Data Ingestion



Data Validation



Data Cleaning



Processed Data



SQLite Database



Python Analytics



Power BI Dashboard



Business Insights

7. DATA CLEANING AND PREPARATION

Data quality is an important part of the project because the analytical results depend on the accuracy and consistency of the underlying datasets.

The Fund Master dataset was inspected for missing values and duplicate records.

The analysis identified:

- **40 records**
- **15 columns**
- **No missing values**
- **No duplicate records**

Date fields were converted into standard datetime format. Numeric fields such as:

- NAV
- AUM
- expense ratio
- SIP amount
- transaction amount
- portfolio weight

were converted into suitable numeric data types. For NAV analysis, records were sorted by AMFI code and date before calculating daily percentage returns.

The first observation for each scheme naturally has no previous NAV and therefore produces a missing daily return. These initial missing return values were excluded from return-based calculations.

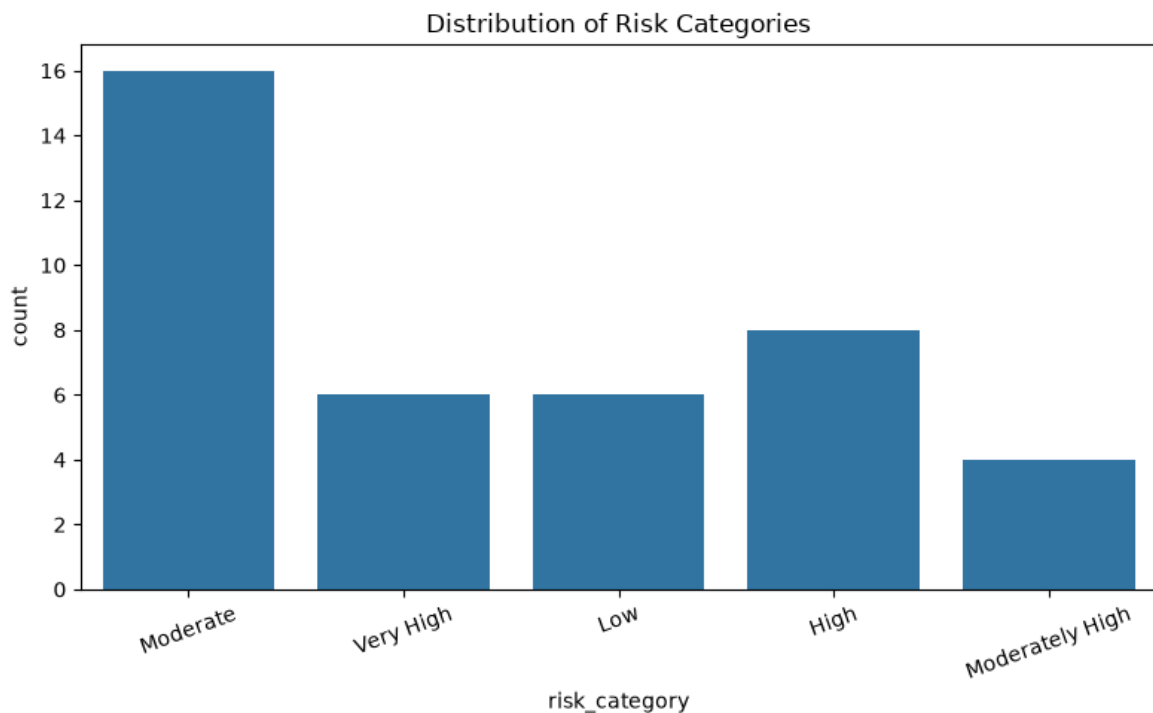
Investor transaction data was cleaned and prepared for cohort and SIP continuity analysis.

Portfolio holdings were prepared so that sector weights could be used to calculate concentration.

8. EXPLORATORY DATA ANALYSIS

8.1 Fund Universe

The Fund Master dataset contains **40 mutual fund schemes across 10 fund houses**. The analysed dataset includes both Equity and Debt categories. Equity schemes form a larger portion of the analysed fund universe.

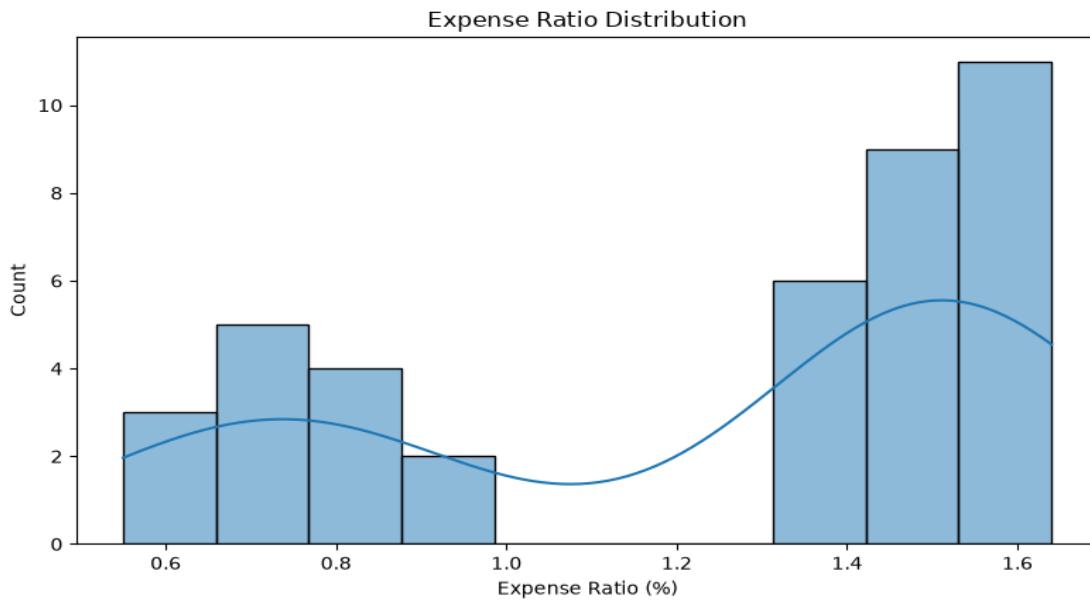


The risk-category distribution shows that Moderate-risk schemes represent the largest group within the analysed fund universe. The presence of Low, Moderate, High and Very High risk categories demonstrates the different levels of risk available to investors.

8.2 Expense Ratio Analysis

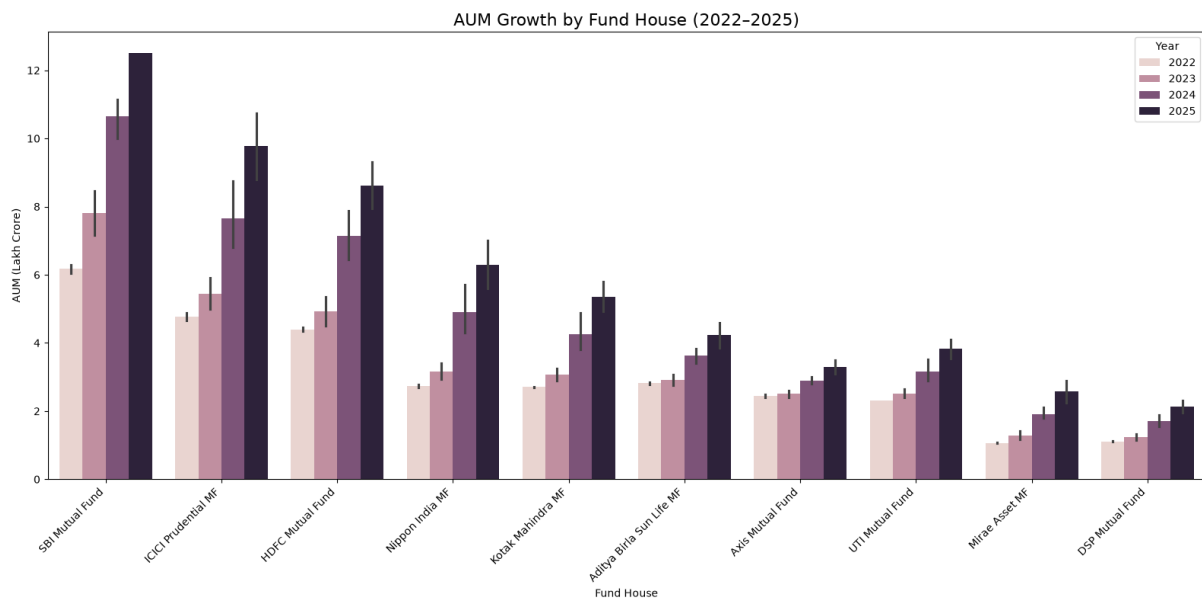
The average expense ratio across the analysed schemes is approximately : **1.237%**

The maximum observed expense ratio is : **1.64%**



Expense ratios represent the cost charged by mutual funds for managing the portfolio. Differences in expense ratios can affect the net returns received by investors over longer periods.

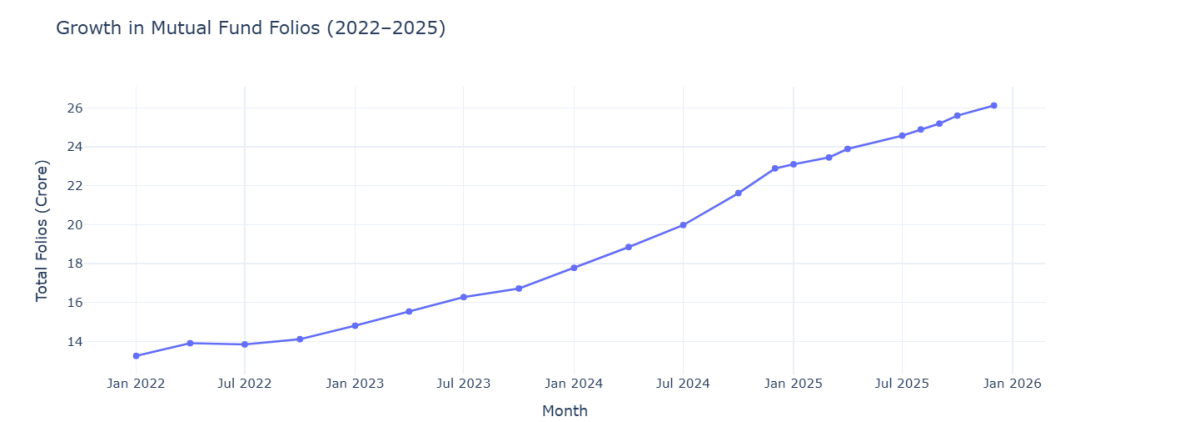
8.3 Assets Under Management



The AUM analysis provides an overview of how assets under management changed across fund houses during the analysed period. AUM growth can indicate changes in investor participation, market performance and net investment flows.

8.4 Folio Growth

Growth in Mutual Fund Folios (2022–2025)

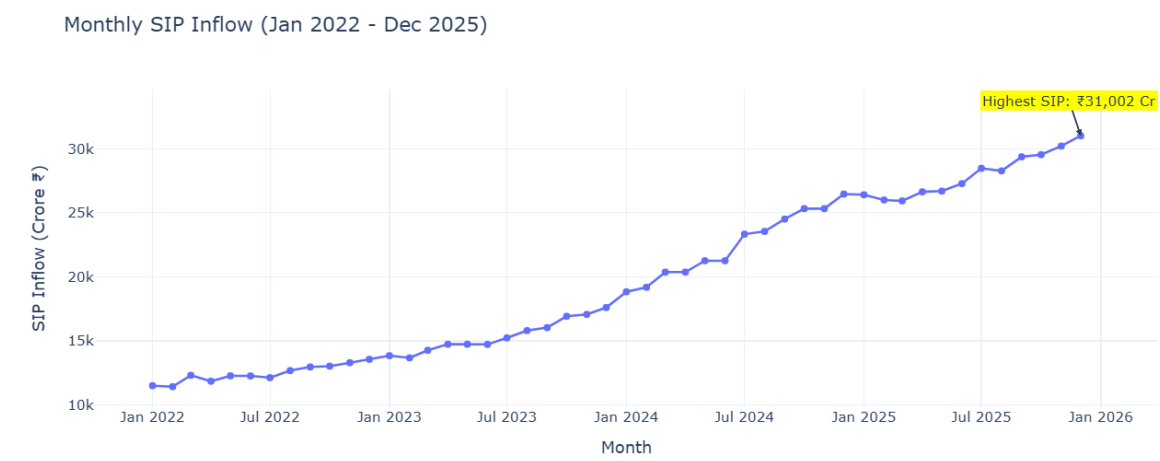


The growth in folios indicates changes in investor participation over time. Increasing folio counts can indicate broader participation in mutual fund investments.

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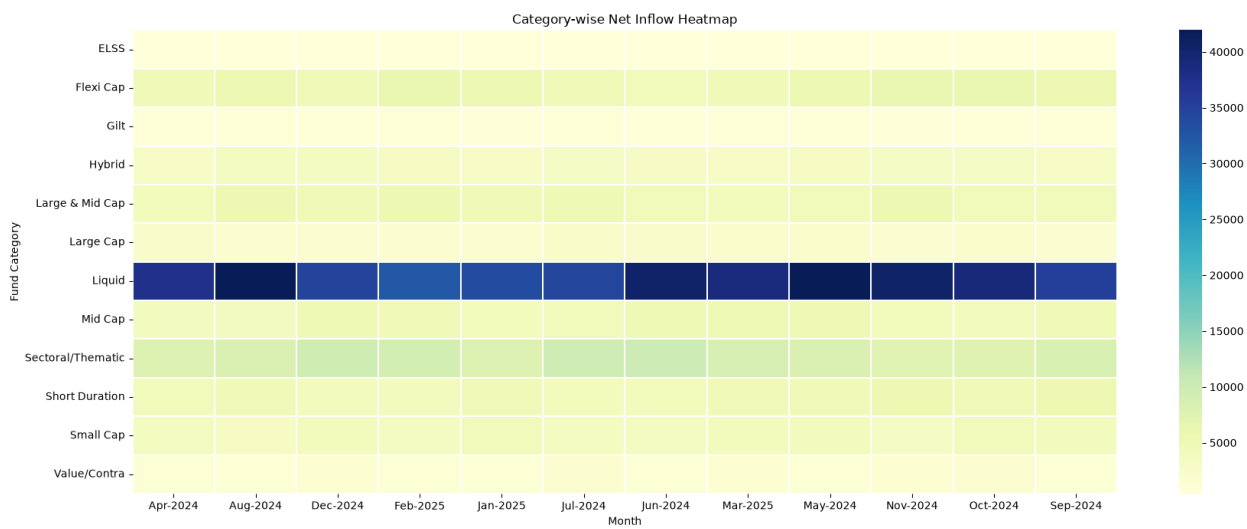
9.SIP AND CATEGORY INFLOWS ANALYSIS

9.1 Monthly SIP Inflow



The SIP inflow analysis provides a time-series view of systematic investment activity. It helps identify changes in investor participation and investment behaviour over the analysed period.

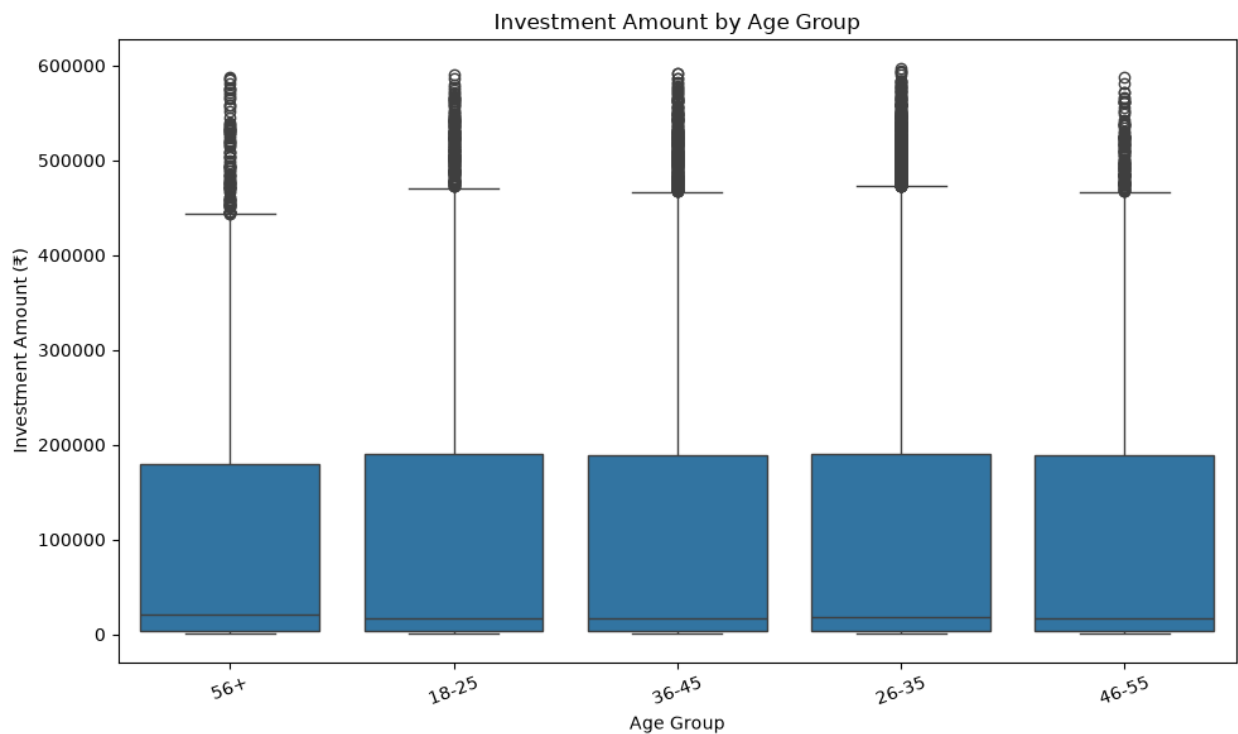
9.2 Category-wise Inflows(Category-wise Net Inflow Heatmap)



The category-level inflow analysis highlights differences in investment preferences across mutual fund categories. The heatmap provides a useful way to identify periods and categories with relatively higher or lower net inflows.

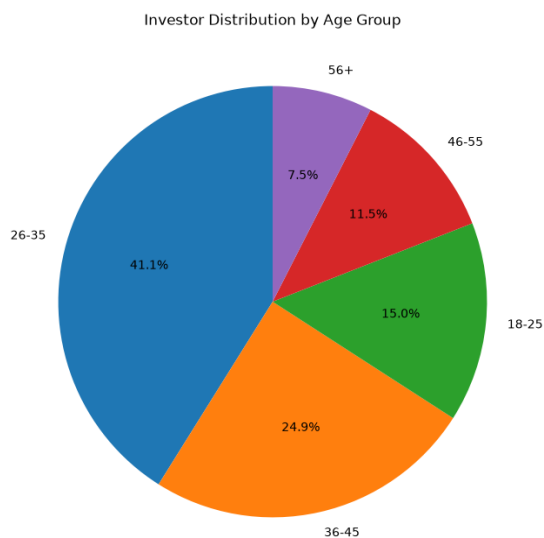
10. INVESTOR ANALYSIS

10.1 Investment Amount by Age Group



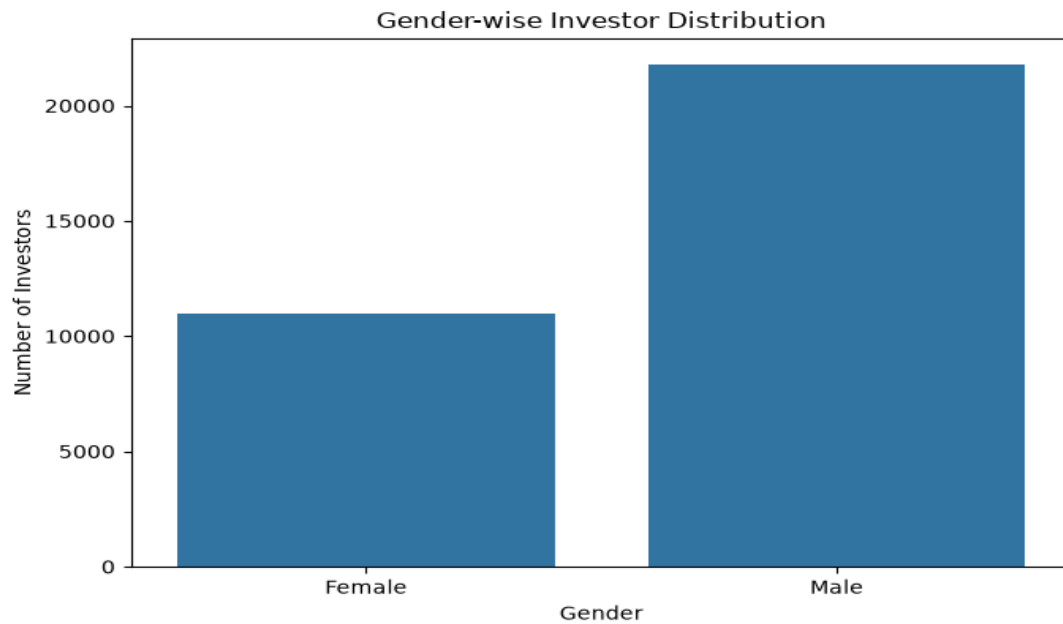
Investment amounts vary across age groups, indicating differences in investment capacity and financial participation.

10.2 Investor Distribution by Age



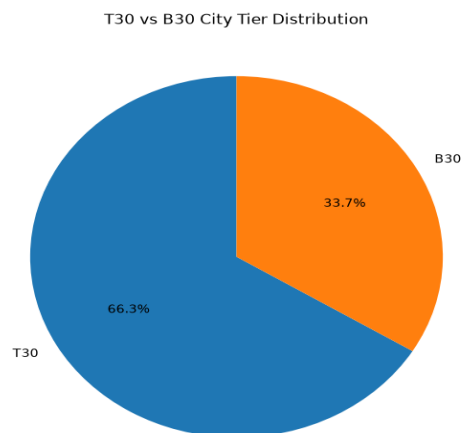
The age distribution provides an overview of the demographic structure of the investor dataset.

10.3 Gender-wise Distribution



The gender distribution provides additional demographic context for understanding the investor population represented in the dataset.

10.4 T30 and B30 Distribution



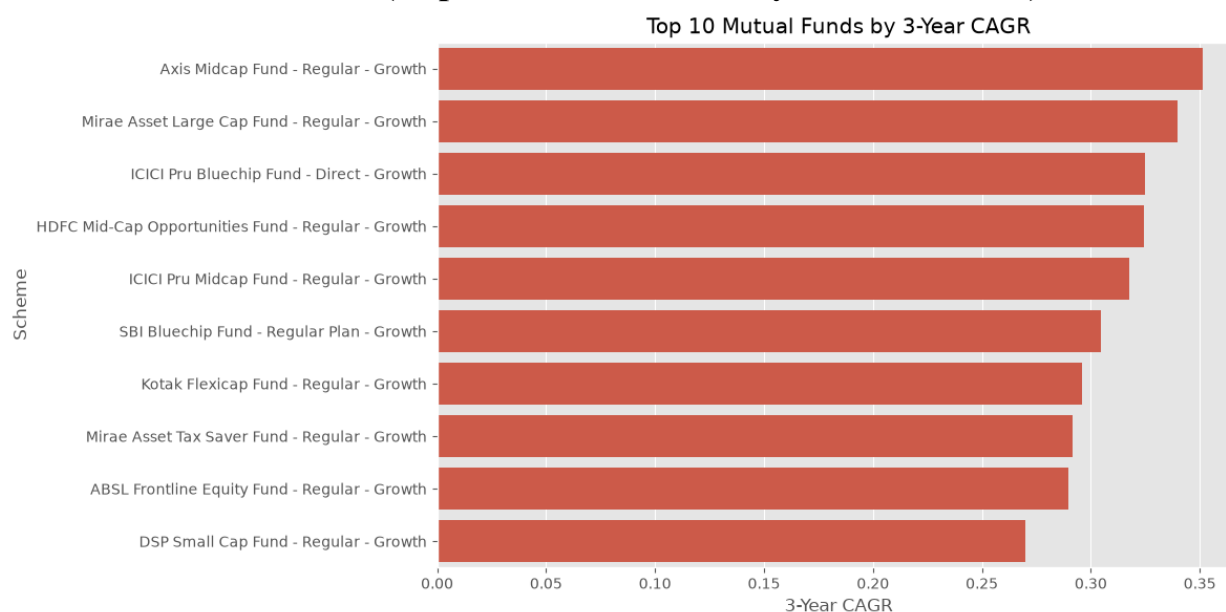
The T30 and B30 analysis provides an indication of mutual fund participation across different city classifications

11. MUTUAL FUND PERFORMANCE ANALYSIS

Historical NAV data was used to calculate daily returns for the analysed schemes. Performance analysis considered several measures including:

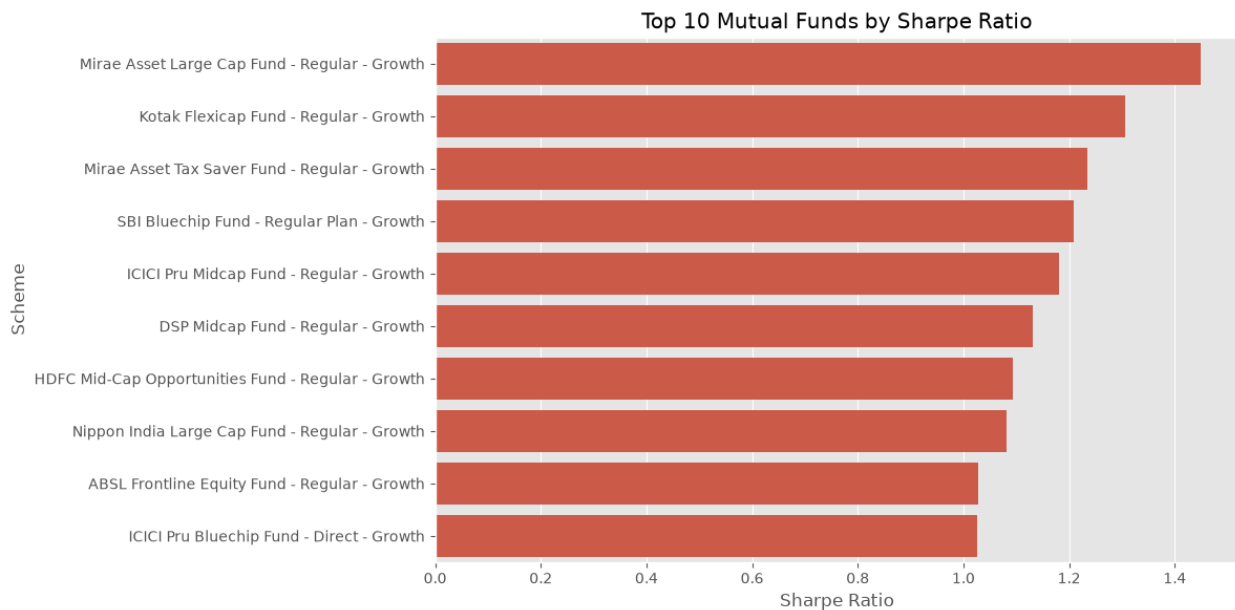
- Daily returns
- CAGR
- Sharpe ratio
- Alpha
- Beta
- Maximum drawdown
- Tracking error
- Benchmark comparison

11.1 Three-Year CAGR(Top 10 Mutual Funds by 3-Year CAGR)



Three-year CAGR provides a longer-term perspective on fund growth. It is useful for comparing the historical growth rates of schemes over a common investment horizon.

11.2 Sharpe Ratio



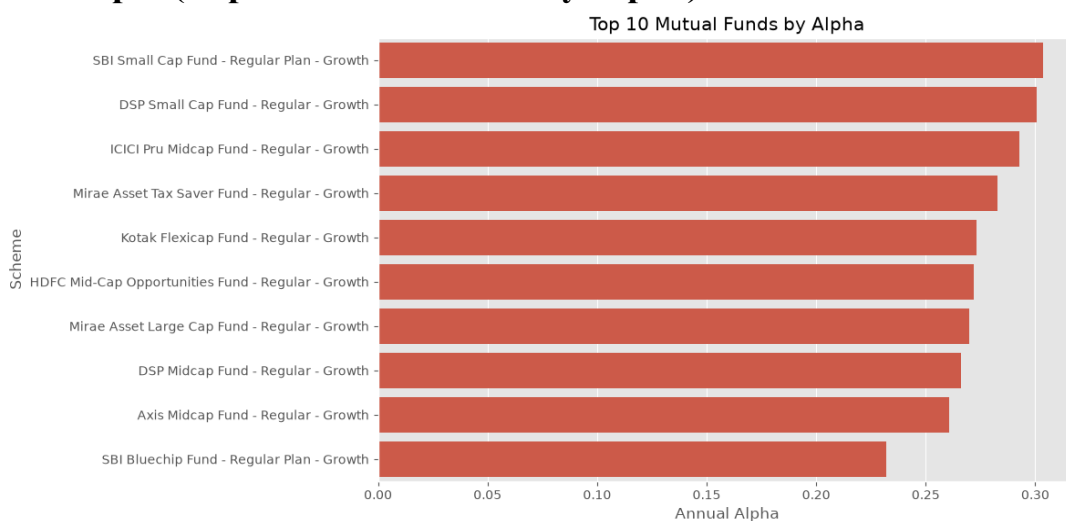
The Sharpe ratio measures historical return relative to volatility. A higher Sharpe ratio generally indicates better historical risk-adjusted performance.

The highest historical Sharpe ratio in the advanced analysis was:

ICICI Pru Liquid Fund - Regular - Growth

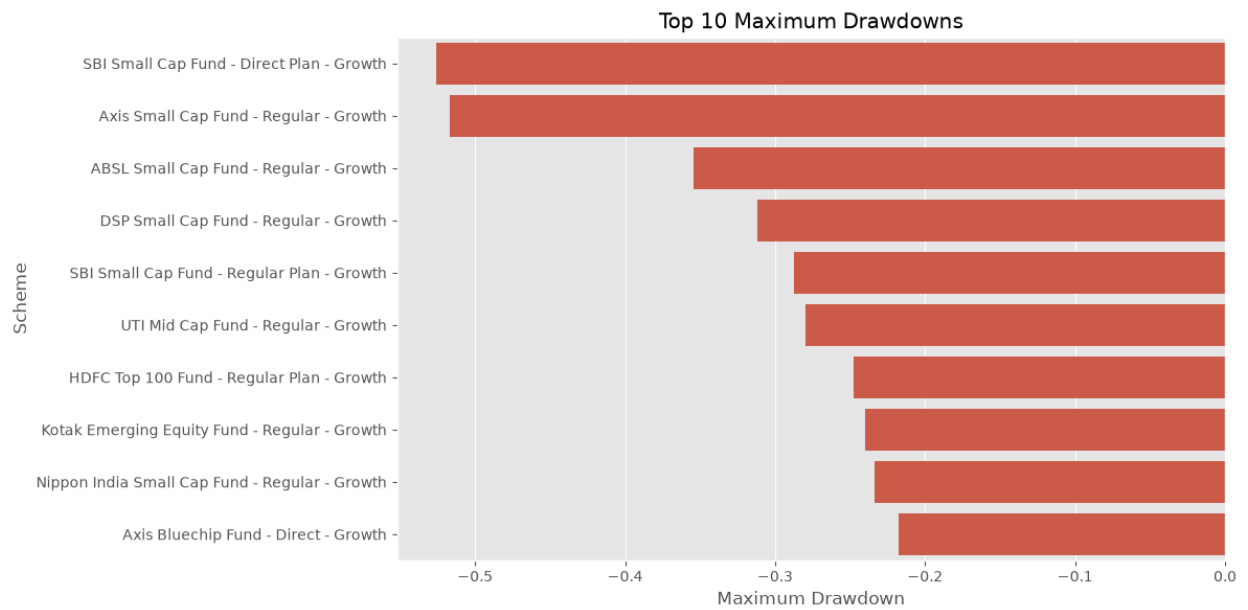
Sharpe ratio: **13.6559**. This result should be interpreted within the fund category because liquid funds generally exhibit substantially lower volatility than equity funds.

11.3 Alpha(Top 10 Mutual Funds by Alpha)



Alpha measures the excess return of a fund relative to its benchmark under the selected analytical methodology.

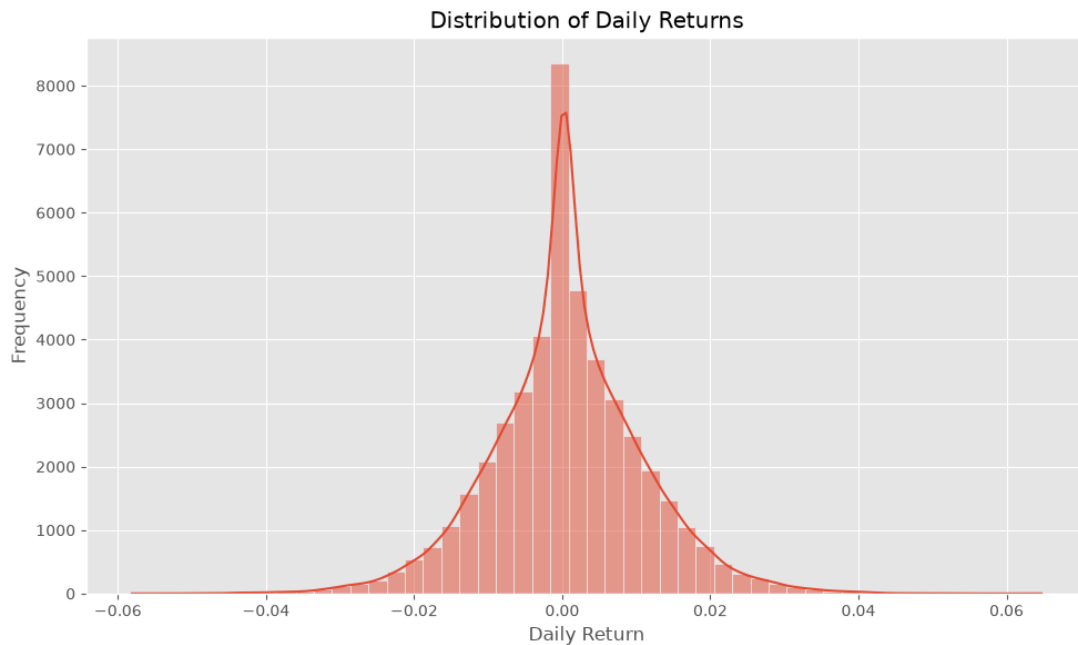
11.4 Maximum Drawdown(Top 10 Maximum Drawdowns)



Maximum drawdown measures the largest historical decline from a peak to a subsequent trough. It provides an additional perspective on downside experience.

12. RISK ANALYSIS

12.1 Daily Return Distribution



The distribution of daily returns helps understand the variability and shape of historical fund returns.

12.2 Historical VaR

Historical Value at Risk at the 95% confidence level was calculated using the 5th percentile of the daily return distribution. A more negative VaR indicates greater historical downside exposure. The analysis identified:

SBI Small Cap Fund - Direct Plan - Growth

with a 95% VaR of approximately - **2.69%**

This indicates that the lower tail of its historical daily return distribution contained relatively larger negative observations compared with the other analysed schemes.

12.3 Conditional VaR

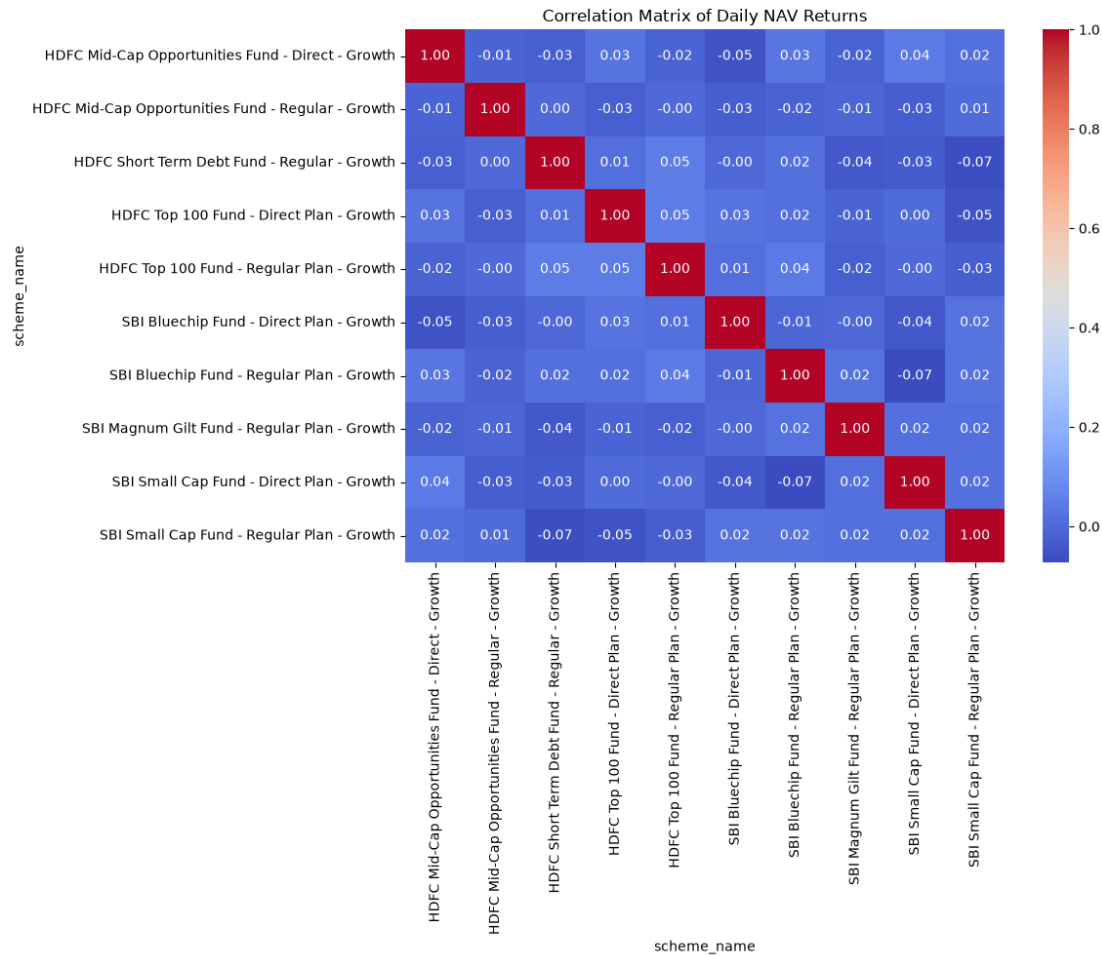
Conditional Value at Risk was calculated as the mean of returns falling below the VaR threshold. The highest downside tail risk was observed for:

ABSL Small Cap Fund - Regular - Growth

with a CVaR of approximately - **3.25%**

CVaR provides additional information because it measures the average severity of losses beyond the VaR threshold.

12.4 Correlation Analysis(Correlation Matrix of Daily NAV Returns)



The correlation matrix helps identify relationships between fund return series. Funds with lower correlations may provide greater diversification benefits when considered together.

13. ADVANCED ANALYTICS

The advanced analytics stage extends the project beyond basic descriptive analysis.

It includes:

- Historical VaR
- CVaR
- Rolling Sharpe
- Investor cohorts
- SIP continuity
- Fund recommender
- HHI concentration

14. INVESTOR COHORT ANALYSIS

Investors were grouped according to the year of their first transaction.

The analysis calculated:

- Average SIP amount
- Total invested amount
- Top fund preference

The **2024 investor cohort** recorded the highest total investment.

Total invested : **₹21.50 crore approximately**

The **2025 cohort** recorded the highest average SIP contribution.

Average SIP : **₹13,505.21**

The results show that the cohort contributing the highest total investment is not necessarily the same cohort with the highest average SIP contribution. This indicates differences in the number of investors, transaction frequency and individual contribution sizes across cohorts.

15. SIP CONTINUITY ANALYSIS

SIP continuity was evaluated for investors with at least six SIP transactions. A gap greater than 35 days between SIP contributions was used to identify investors as at-risk. The analysis identified:

Eligible investors: 1,362

Regular investors: 30

At-risk investors: 1,332

Therefore:

Regular SIP rate: 2.2%

At-risk rate: 97.8%

The high proportion of investors classified as at-risk indicates that SIP contribution behaviour is highly irregular under the selected 35-day threshold. This can be useful for investor retention analysis and targeted engagement strategies.

16. SIMPLE FUND RECOMMENDER

A simple rule-based recommender was developed using investor risk appetite.

The user can enter:

- Low
- Moderate
- High

The system filters funds according to their risk category and ranks the matching funds by Sharpe ratio.

For example, the Low-risk recommendation included:

1. ICICI Pru Liquid Fund - Regular - Growth
2. Kotak Liquid Fund - Regular - Growth
3. ABSL Liquid Fund - Regular - Growth

For Moderate risk, the top recommendations included:

1. Mirae Asset Large Cap Fund - Regular - Growth
2. SBI Bluechip Fund - Regular Plan - Growth
3. Nippon India Large Cap Fund - Regular - Growth

For High risk, the top recommendations included:

1. Mirae Asset Tax Saver Fund - Regular - Growth
2. ICICI Pru Midcap Fund - Regular - Growth
3. DSP Midcap Fund - Regular - Growth

The recommender is intended as an analytical demonstration and not as personalized financial advice.

17. PORTFOLIO SECTOR CONCENTRATION — HHI

The Herfindahl-Hirschman Index was used to measure sector concentration.

The formula is : **HHI = $\Sigma(\text{weight}_i^2)$**

A higher HHI indicates greater concentration in fewer sectors.

A lower HHI indicates greater diversification.

The most concentrated fund was:

Axis Bluechip Fund - Regular - Growth

HHI : 0.2968

The most diversified fund in the analysis was:

UTI Mid Cap Fund - Regular - Growth

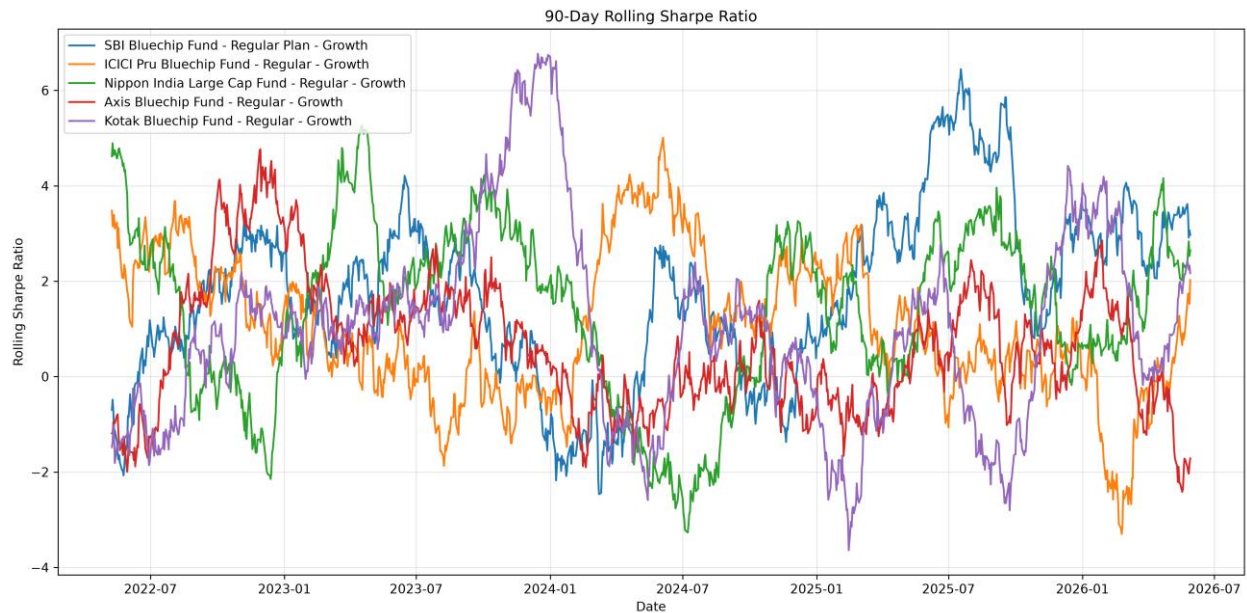
HHI : 0.1240. The results indicate meaningful differences in sector concentration across funds. Funds with higher HHI values may have greater exposure to a smaller number of sectors.

18. ROLLING 90-DAY SHARPE ANALYSIS

Rolling 90-day Sharpe ratios were calculated using:

- 90-day rolling mean returns
- 90-day rolling standard deviation
- Annualization using $\sqrt{252}$

The rolling analysis provides a time-varying view of risk-adjusted performance.

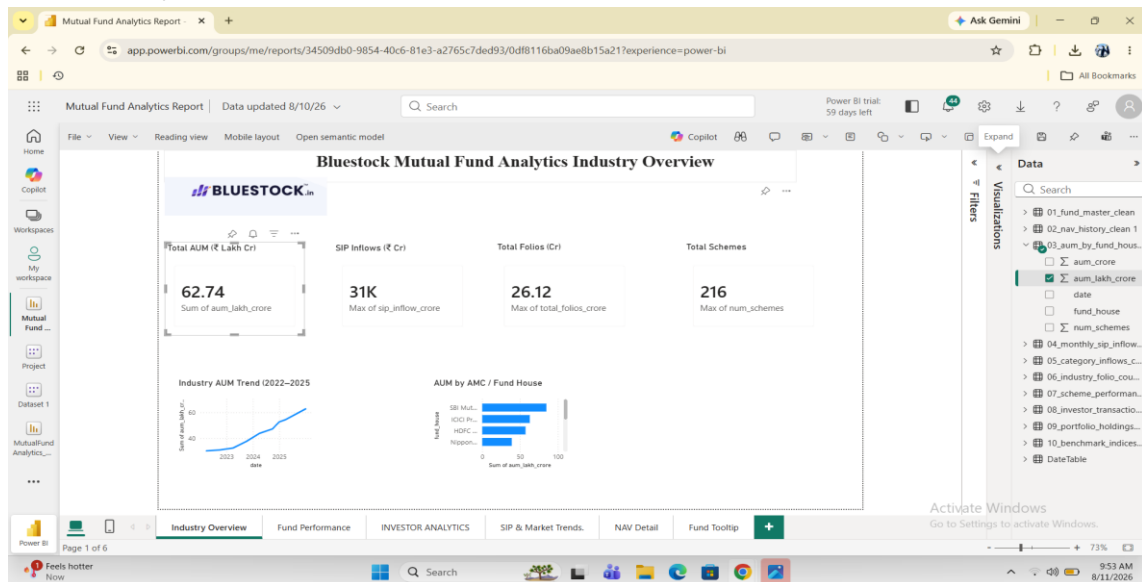


Unlike a single historical Sharpe ratio, the rolling Sharpe measure allows changes in risk-adjusted performance to be observed over time. Periods with higher rolling Sharpe values indicate stronger historical return relative to recent volatility, while lower values indicate weaker risk-adjusted performance.

19. INTERACTIVE POWER BI DASHBOARD

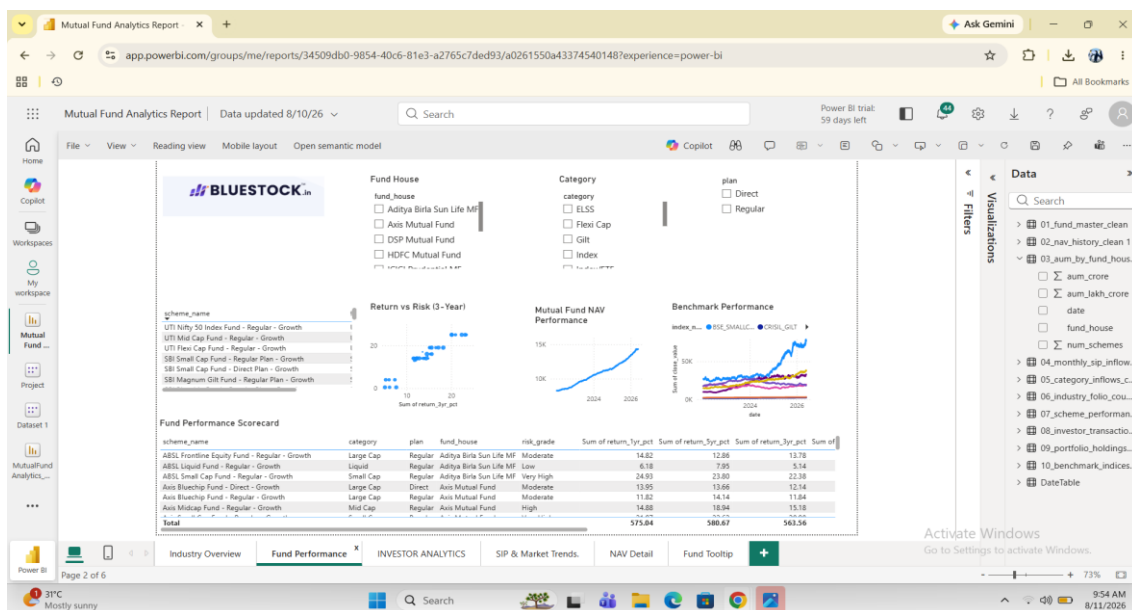
The final Power BI solution consists of four primary analytical pages and one NAV Detail drill-through page. The dashboard includes interactive filters, KPI cards, charts and drill-through functionality.

19.1 Industry Overview



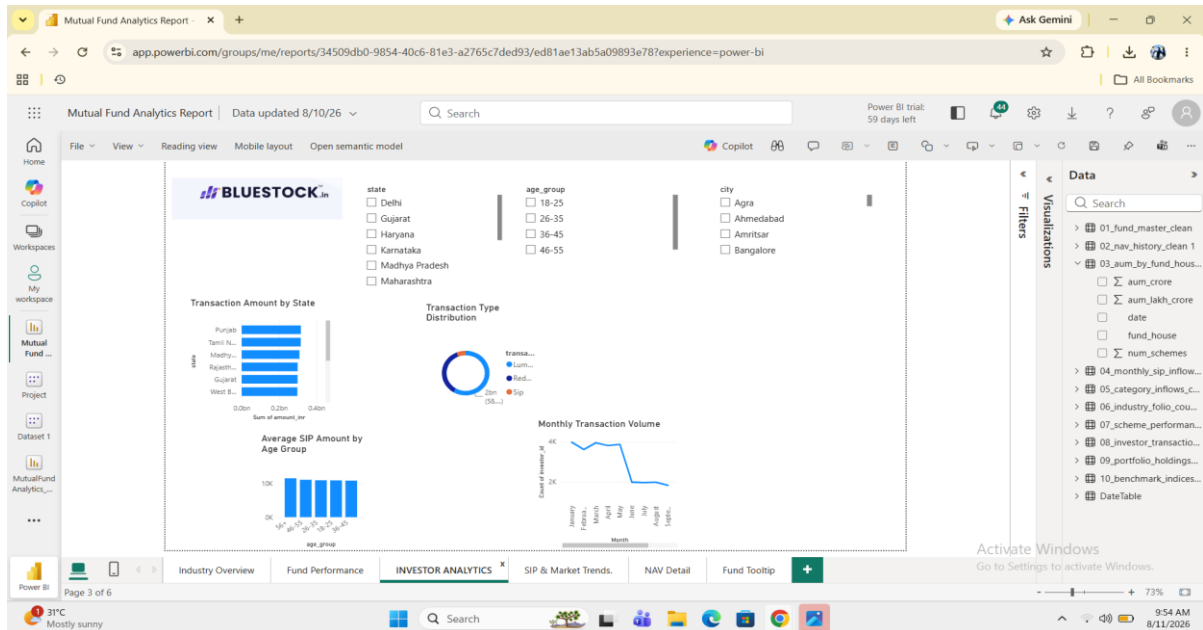
The Industry Overview page provides a high-level view of mutual fund industry metrics, fund-house information and market-level indicators. It is designed to provide users with a quick overview before moving into detailed fund or investor analysis.

19.2 Fund Performance



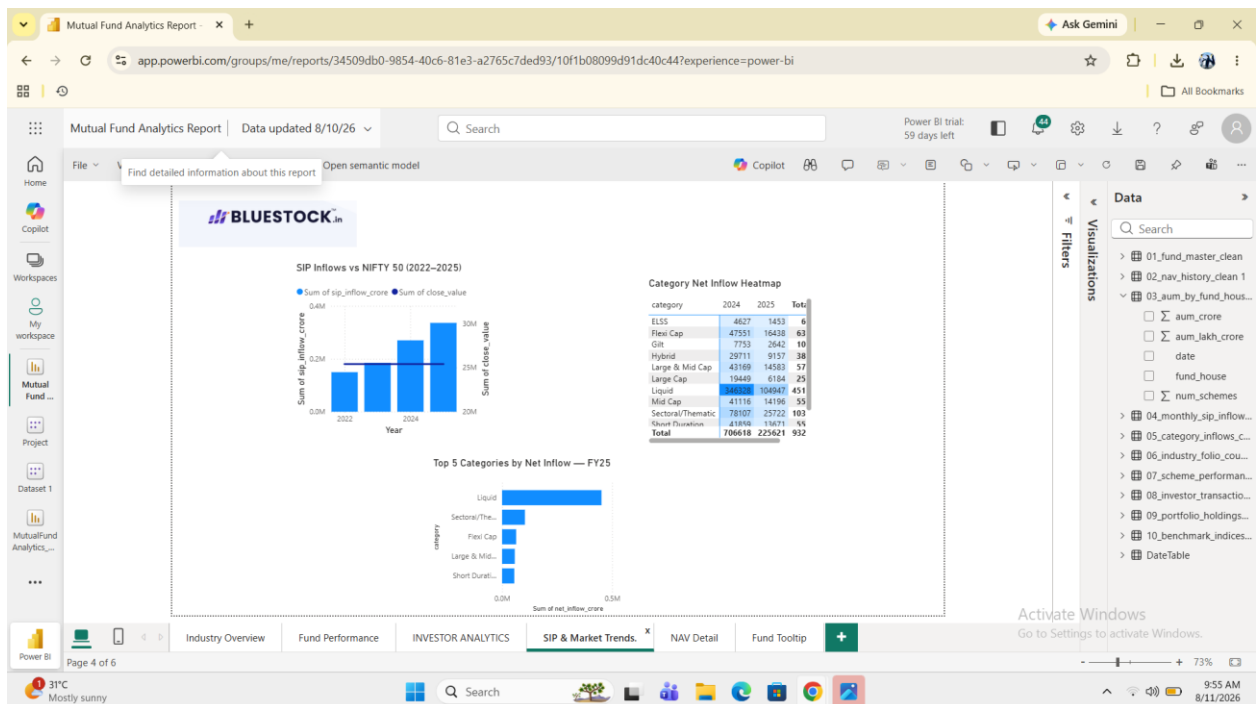
The Fund Performance page focuses on mutual fund performance indicators and enables users to compare schemes using relevant performance measures.

19.3 Investor Analytics



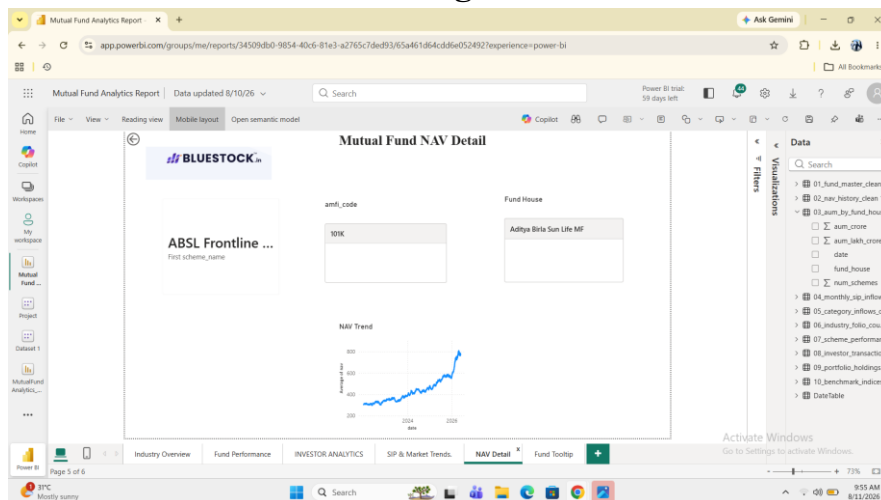
The Investor Analytics page presents demographic and transaction-related insights, helping users understand investor characteristics and investment behaviour.

19.4 SIP and Market Trends



The SIP and Market Trends page focuses on SIP activity, market trends and related investment-flow indicators.

19.5 NAV Detail Drill-through



The NAV Detail page is implemented as a drill-through page. Users can select a particular scheme from the dashboard and navigate to the NAV Detail page to examine its historical NAV trend and scheme-level information. This page is treated as a drill-through page in addition to the four primary dashboard pages.

20. KEY FINDINGS

The major findings from the project are summarized below.

Finding 1 — Downside Risk

SBI Small Cap Fund - Direct Plan - Growth recorded the most negative historical VaR at approximately **-2.69%**.

ABSL Small Cap Fund - Regular - Growth recorded the most negative CVaR at approximately **-3.25%**.

Finding 2 — Investor Cohorts

The 2024 investor cohort recorded the highest total investment at approximately **₹21.50 crore**.

The 2025 cohort recorded the highest average SIP contribution at approximately **₹13,505.21**.

Finding 3 — SIP Continuity

Only **2.2%** of eligible investors were classified as regular based on the selected SIP continuity rule. Approximately **97.8%** were classified as at-risk.

Finding 4 — Risk-adjusted Performance

ICICI Pru Liquid Fund - Regular - Growth recorded the highest historical Sharpe ratio at approximately **13.66**. The result should be interpreted within the fund's category because volatility differs substantially between liquid and equity funds.

Finding 5 — Portfolio Concentration

Axis Bluechip Fund - Regular - Growth had the highest HHI at **0.2968**, indicating comparatively greater sector concentration. UTI Mid Cap Fund - Regular - Growth had the lowest HHI at **0.1240**, indicating greater diversification.

21. RECOMMENDATIONS

Based on the analysis, the following recommendations can be considered.

1. Consider risk alongside returns

Investors should not evaluate funds only on historical returns. Measures such as Sharpe ratio, VaR, CVaR and maximum drawdown provide additional information about risk.

2. Monitor downside exposure

Funds with more negative VaR and CVaR values should be examined carefully because their historical return distributions indicate greater downside exposure.

3. Monitor SIP continuity

The high proportion of at-risk investors suggests that SIP continuity can be an important area for investor engagement. Automated reminders and targeted communication could potentially improve SIP continuity.

4. Consider diversification

Funds with high HHI values should be monitored for sector concentration. Investors may consider diversification across funds with different sector exposures.

5. Improve the recommender

The current recommender is based primarily on risk category and Sharpe ratio. A future version could include:

- CAGR
- maximum drawdown
- expense ratio
- Alpha, Beta
- fund age, Category
- portfolio concentration

6. Use the dashboard for monitoring

The Power BI dashboard can be used as a monitoring interface for fund performance, SIP activity, investor behaviour and NAV trends.

22. LIMITATIONS

The project has several limitations.

1. Historical performance does not guarantee future performance.
2. Historical VaR and CVaR are dependent on the historical return distribution and may not capture unexpected market events.
3. Sharpe ratio results depend on the selected time period and volatility characteristics.
4. Investor analysis is limited to the available transaction dataset.
5. SIP continuity depends on transaction dates available in the dataset.
6. HHI measures sector concentration but does not capture all dimensions of portfolio risk.
7. The fund recommender is a simple rule-based analytical model and should not be interpreted as personalized investment advice.
8. External market conditions and macroeconomic factors are not fully incorporated into the recommendation model.

23. CONCLUSION

The Bluestock Mutual Fund Analytics project demonstrates how multiple mutual fund datasets can be transformed into a structured analytical solution.

The project combines data ingestion, data cleaning, exploratory analysis, performance measurement, risk analytics, investor behaviour analysis, portfolio concentration analysis and interactive visualization.

The analysis of 40 mutual fund schemes across 10 fund houses provides insights into historical performance, downside risk, investor contribution patterns and portfolio diversification.

The advanced analytics further demonstrate how VaR, CVaR, rolling Sharpe ratio, investor cohorts, SIP continuity, fund recommendation and HHI can be combined to provide a broader understanding of mutual fund behaviour.

The Power BI dashboard makes the results accessible through interactive visualizations and drill-through functionality.

Overall, the project demonstrates the practical application of Python, Pandas, NumPy, SQL, SQLite, statistical analysis and Power BI in financial data analytics.